Examen Final - G.T.

(1)

$$S = \frac{13+9+10}{2} = 16$$

$$A = \sqrt{s(s-a)(s-b)(s-c)}$$

$$A = \sqrt{16(16-13)(16-9)(16-10)}$$

$$A = \sqrt{16 \cdot 3 \cdot 7 \cdot 6} = \underline{12\sqrt{14}}$$

(2)

$$5\,\operatorname{sen}^2 x + \cos^2 x = 2$$

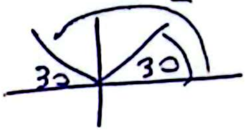
$$5\,\operatorname{sen}^2 x + (1 - \operatorname{sen}^2 x) = 2$$

$$4\,\operatorname{sen}^2 x = 1$$

$$\operatorname{sen}^2 x = \frac{1}{4}$$

$$\operatorname{sen} x = \pm \frac{1}{2}$$

*) $\operatorname{sen} x = \frac{1}{2} \Rightarrow x = \frac{\pi}{6}, \frac{5\pi}{6}$



*) $\operatorname{sen} x = -\frac{1}{2} \Rightarrow x = \frac{7\pi}{6}, \frac{11\pi}{6}$

La suma es:

$$\text{Suma} = \frac{\pi}{6} + \frac{5\pi}{6} + \frac{7\pi}{6} + \frac{11\pi}{6} = \underline{4\pi}$$

3

$$Z = \tan\left(\frac{\pi}{4} + \alpha\right) + \tan\left(\frac{\pi}{4} - \alpha\right)$$

$$= \frac{\tan\frac{\pi}{4} + \tan\alpha}{1 - \tan\frac{\pi}{4}\tan\alpha} + \frac{\tan\frac{\pi}{4} - \tan\alpha}{1 + \tan\frac{\pi}{4}\tan\alpha}$$

$$= \frac{1 + \tan\alpha}{1 - \tan\alpha} + \frac{1 - \tan\alpha}{1 + \tan\alpha}$$

$$= \frac{(1+\tan\alpha)^2 + (1-\tan\alpha)^2}{(1-\tan\alpha)(1+\tan\alpha)} =$$

$$= \frac{(1 + 2\tan\alpha + \tan^2\alpha) + (1 - 2\tan\alpha + \tan^2\alpha)}{1 - \tan^2\alpha}$$

$$= \frac{2(1 + \tan^2\alpha)}{1 - \tan^2\alpha} = \frac{2\left(1 + \frac{\sin^2\alpha}{\cos^2\alpha}\right)}{1 - \frac{\sin^2\alpha}{\cos^2\alpha}}$$

$$= \frac{2 \cdot \frac{\cos^2\alpha + \sin^2\alpha}{\cos^2\alpha}}{\frac{\cos^2\alpha - \sin^2\alpha}{\cos^2\alpha}} = \frac{2}{\cos^2\alpha - \sin^2\alpha}$$

$$= \frac{2}{\cos 2\alpha} = \underline{\underline{2\sec(2\alpha)}}$$

(4)

$$x^2 - 8x + 6y + 4 = 0$$

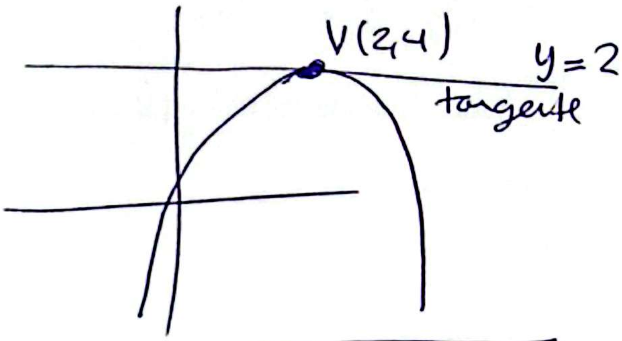
$$x^2 - 8x = -6y - 4$$

$$x^2 - 8x + 4^2 = -6y - 4 + 4^2$$

$$(x-4)^2 = -6y + 12$$

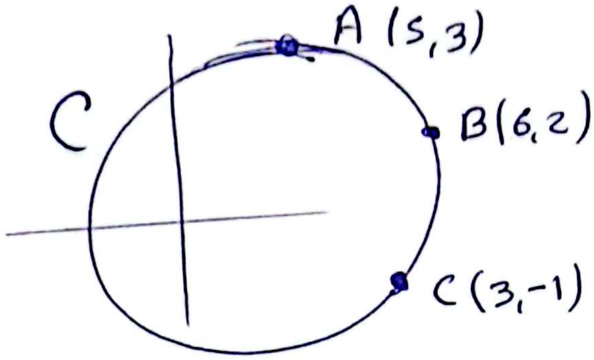
$$(x-4)^2 = -6(y-2) \Rightarrow V(4,2)$$

La tangente es en el vertice, entonces



Sol. $\boxed{y - 2 = 0}$

(5)



La ecuación es: $x^2+y^2+Dx+Ey+F=0$

$A(5,3) \in C \Rightarrow 5^2+3^2+5D+3E+F=0$
$B(6,2) \in C \Rightarrow 6^2+2^2+6D+2E+F=0$
$C(3,-1) \in C \Rightarrow 3^2+(-1)^2+3D-E+F=0$

luego
$$\begin{cases} 5D+3E+F=-34 \\ 6D+2E+F=-40 \\ 3D-E+F=-10 \end{cases}$$

Sol. $D=-8$, $E=-2$, $F=12$

La circunferencia es:

$$x^2+y^2-8x-2y+12=0$$
$$x^2-8x \quad +y^2-2y \quad =-12$$
$$(x^2-8x+4^2)+(y^2-2y+1^2)=-12+4^2+1^2$$
$$(x-4)^2+(y-1)^2=5$$
$$(x-4)^2+(y-1)^2=\sqrt{5}^2$$

$\therefore$ $\boxed{R=\sqrt{5}}$